

INTERNSHIP REPORT

ON

WEB DEVELOPMENT INTERNSHIP

Project: SMS Spam Detection Using Machine Learning

**Submitted in partial fulfillment of the requirements for the degree of
Bachelor of Technology in Computer Science and Engineering**

**Internship Organization:
Codec Technologies Pvt. Ltd.**

**Internship Role:
Web Development Intern**

**Internship Duration:
01 July 2026 – 30 July 2026**

**Internship Mode:
Pan India, Hybrid**

**Internship Type:
AICTE & ICAC Approved**

DECLARATION

I hereby declare that this internship report titled “**Web Development Internship at Codec Technologies Pvt. Ltd.**”, including the project “**SMS Spam Detection Using Machine Learning**”, is an original record of the work and learning completed by me during the internship period from **01 July 2026 to 30 July 2026**.

The internship was undertaken in the role of **Web Development Intern** at **Codec Technologies Pvt. Ltd.**. The work presented in this report reflects my understanding, practical learning, technical activities, and project experience gained during the internship.

I further declare that this report has been prepared for academic purposes and has not been submitted elsewhere for the award of any other degree or qualification.

Name: ____

Course: B.Tech Computer Science and Engineering

Session: 2025–2029

Date: ____

Signature: _____

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Codec Technologies Pvt. Ltd.** for providing me with the opportunity to undertake a one-month internship as a **Web Development Intern**.

This internship provided me with valuable exposure to a professional working environment and helped me understand how theoretical concepts are applied to practical projects. During the internship, I gained experience in programming, data handling, problem-solving, project development, debugging, documentation, and understanding the overall software development workflow.

I am especially thankful to the project head and mentors for their guidance, support, and feedback throughout the internship. Their suggestions helped me improve my technical understanding and approach problems in a more structured manner.

I am also grateful to my faculty members and institution for encouraging practical learning and providing the academic support required to complete this internship successfully.

Finally, I would like to thank everyone who directly or indirectly contributed to the successful completion of my internship and this report.

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CHAPTER 1: INTRODUCTION

1.1 Introduction to the Internship

An internship provides students with an opportunity to gain practical exposure beyond classroom learning. It helps bridge the gap between theoretical knowledge and real-world implementation by allowing students to work on practical tasks and understand professional development processes.

I completed a one-month internship at **Codec Technologies Pvt. Ltd.** from **01 July 2026 to 30 July 2026** as a **Web Development Intern.**

The internship was conducted in a **Pan India, Hybrid** mode and was approved under **AICTE and ICAC**. During the internship, I was introduced to practical software development practices and worked on a project titled **"SMS Spam Detection Using Machine Learning."**

Although the project involved Machine Learning and Natural Language Processing concepts, the internship provided broader exposure to development practices, Python programming, data processing, debugging, documentation, and systematic project implementation.

1.2 Importance of the Internship

The internship helped me understand how software projects are developed and managed in a professional environment. It provided an opportunity to:

- Apply programming concepts to practical problems.
 - Understand project requirements.
 - Work with real-world datasets.
 - Develop structured solutions.
 - Debug and improve implementations.
 - Compare different technical approaches.
 - Understand the importance of testing and evaluation.
 - Improve technical documentation.
 - Develop professional communication and problem-solving skills.
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CHAPTER 2: ORGANIZATION PROFILE

2.1 About Codec Technologies Pvt. Ltd.

Codec Technologies Pvt. Ltd. provided the internship environment in which I gained practical exposure to software and technology-related activities.

The internship focused on practical learning and project-based development. The organization provided guidance and a structured environment through which I was able to understand different stages of project development.

2.2 Internship Role

My designated role during the internship was:

Web Development Intern

As a Web Development Intern, I was exposed to software development practices, programming, problem-solving, project implementation, and documentation.

The assigned project also introduced me to Machine Learning-based development through the implementation of an SMS spam classification system.

2.3 Internship Duration

The internship was completed over a period of one month:

01 July 2026 – 30 July 2026

2.4 Mode of Internship

The internship was conducted in:

Pan India, Hybrid Mode

This mode required independent learning, regular project work, communication, and systematic completion of assigned tasks.

CHAPTER 3: INTERNSHIP OVERVIEW

3.1 Internship Details

Particular	Details
Organization	Codec Technologies Pvt. Ltd.
Role	Web Development Intern
Duration	01 July 2026 – 30 July 2026
Mode	Pan India, Hybrid
Type	AICTE & ICAC Approved
Project	SMS Spam Detection Using Machine Learning
Reporting	Assigned Project Head(s)

3.2 Nature of the Internship

The internship was primarily focused on practical software development and project-based learning.

The experience involved understanding requirements, writing and executing Python programs, handling data, preprocessing text, implementing Machine Learning algorithms, evaluating results, debugging errors, and documenting the completed work.

The internship also helped me understand that successful software development requires more than simply writing code. It involves planning, testing, debugging, evaluation, documentation, and continuous improvement.

CHAPTER 4: OBJECTIVES OF THE INTERNSHIP

The major objectives of the internship were:

4.1 Technical Objectives

- Develop practical programming skills.
- Improve understanding of Python.
- Understand software development workflows.
- Work with real-world datasets.
- Learn data preprocessing techniques.
- Understand text classification.
- Implement Machine Learning algorithms.
- Evaluate and compare different models.

- Understand feature extraction techniques.
- Develop systematic problem-solving abilities.

4.2 Professional Objectives

- Understand professional development practices.
 - Improve communication skills.
 - Learn to work independently.
 - Develop time-management skills.
 - Improve documentation abilities.
 - Learn how to approach technical problems.
 - Gain exposure to project-based development.
-

CHAPTER 5: TRAINING AND LEARNING DURING INTERNSHIP

The internship provided learning across multiple areas.

5.1 Python Programming

Python was one of the primary technologies used during the project.

The internship helped strengthen my understanding of:

- Variables and data types
- Conditional statements
- Loops
- Functions
- Lists and dictionaries
- File handling
- Libraries and packages
- Data manipulation
- Exception handling
- Object-oriented programming concepts

Python's extensive ecosystem also made it suitable for data processing and Machine Learning implementation.

5.2 Data Handling

Working with the SMS dataset provided practical exposure to data handling.

I learned how to:

- Load datasets.
- Inspect data.
- Identify missing values.
- Detect duplicate records.
- Clean unnecessary information.
- Analyze class distributions.
- Prepare data for model training.

5.3 Problem Solving

A major learning outcome of the internship was improvement in problem-solving ability.

Instead of directly implementing a solution, I learned to divide a problem into smaller stages:

Requirement → Data → Processing → Implementation → Testing → Evaluation → Improvement

This structured approach made the development process easier to understand.

5.4 Debugging

Debugging was an important part of the project.

While implementing different stages, errors could occur because of:

- Incorrect data types.
- Missing values.
- Incorrect library usage.
- Syntax errors.
- Incorrect model parameters.
- Data-processing issues.

Debugging helped me understand the importance of carefully analyzing errors instead of simply modifying code randomly.

5.5 Documentation

Another important learning outcome was technical documentation.

I learned how to document:

- Project objectives.
- Methodology.
- Technologies used.
- Dataset details.

- Implementation steps.
- Results.
- Limitations.
- Future improvements.

CHAPTER 6: TECHNOLOGIES AND TOOLS USED

The following technologies and tools were used during the internship and project.

Technology/Tool	Purpose
Python	Programming and implementation
Pandas	Data manipulation
NumPy	Numerical operations
Scikit-learn	Machine Learning
NLTK	Natural Language Processing
Matplotlib	Data visualization
Seaborn	Exploratory data analysis
TF-IDF	Text feature extraction
Jupyter Notebook	Development and experimentation
Git/GitHub	Version control and documentation

6.1 Python

Python was used as the primary programming language because of its simple syntax and extensive libraries.

6.2 Pandas

Pandas was used for dataset loading, cleaning, filtering, analysis, and manipulation.

6.3 NumPy

NumPy provided numerical operations and supported efficient data processing.

6.4 Scikit-learn

Scikit-learn was used for:

- Dataset splitting.
- Feature extraction.
- Model training.
- Model evaluation.
- Classification algorithms.
- Ensemble learning.

6.5 NLTK

NLTK was used for Natural Language Processing operations such as tokenization, stopwords removal, and stemming.

CHAPTER 7: PROJECT DESCRIPTION

SMS SPAM DETECTION USING MACHINE LEARNING

7.1 Introduction

SMS messages are commonly used for communication, but unwanted promotional and fraudulent messages can create security and privacy concerns.

The objective of the project was to develop a system capable of automatically classifying SMS messages into two categories:

- **Ham** – legitimate messages.
- **Spam** – unwanted or suspicious messages.

The project used Natural Language Processing techniques to convert text messages into numerical representations and Machine Learning algorithms to classify them.

7.2 Project Objective

The main objective was to develop an effective SMS spam classification system capable of accurately distinguishing between legitimate and spam messages.

The project involved the following major stages:

1. Dataset collection.
2. Data cleaning.

3. Text preprocessing.
 4. Exploratory Data Analysis.
 5. Feature extraction.
 6. Machine Learning model development.
 7. Model comparison.
 8. Ensemble learning.
 9. Final evaluation.
-

CHAPTER 8: DATASET AND DATA UNDERSTANDING

8.1 Dataset

The SMS dataset initially contained:

5,572 records

After cleaning and removing duplicate records, the final dataset contained:

5,169 records

The cleaned dataset consisted of:

Category	Number of Messages
Ham	4,516
Spam	653
Total	5,169

8.2 Dataset Distribution

The dataset contained significantly more legitimate messages than spam messages.

This imbalance is important because a classification model should not simply achieve high accuracy by predicting the majority class.

Therefore, multiple evaluation metrics were considered.

8.3 Target Classes

The target variable consisted of two classes:

Ham

A normal and legitimate SMS message.

Spam

An unwanted or potentially fraudulent SMS message.

CHAPTER 9: DATA PREPROCESSING

Data preprocessing was one of the most important stages of the project.

Raw SMS messages contain punctuation, unnecessary characters, different capitalization, stopwords, and other information that may not be useful for classification.

9.1 Text Cleaning

The messages were cleaned to remove unnecessary elements and standardize the text.

9.2 Tokenization

Tokenization involves dividing a sentence into individual words or tokens.

For example:

Input:

"Congratulations! You have won a prize."

Tokens:

"Congratulations", "You", "have", "won", "a", "prize"

9.3 Stopword Removal

Common words that provide limited classification information were removed.

Examples include words such as:

- the
- is
- a
- an
- and
- of

9.4 Stemming

Stemming reduces words to their basic or root forms.

For example:

playing → play

played → play

This reduces variations of words and can improve consistency in text representation.

CHAPTER 10: EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis was performed to understand the characteristics of the dataset.

10.1 Class Distribution

The dataset contained:

- 4,516 Ham messages.
- 653 Spam messages.

This demonstrated the imbalance between the two classes.

10.2 Message Characteristics

The messages were analyzed to identify differences between spam and legitimate messages.

Spam messages commonly contain promotional language, offers, urgency, prizes, financial terms, or calls to action.

Legitimate messages generally represent normal personal, informational, or service-related communication.

10.3 Importance of EDA

EDA helped in:

- Understanding the dataset.
- Identifying class imbalance.
- Understanding text characteristics.
- Identifying useful patterns.

- Preparing the data for Machine Learning.
-

CHAPTER 11: FEATURE EXTRACTION USING TF-IDF

Machine Learning models cannot directly process raw text.

Therefore, the text messages needed to be converted into numerical features.

11.1 TF-IDF

TF-IDF stands for **Term Frequency-Inverse Document Frequency**.

It assigns numerical importance to words based on their occurrence in a document and their frequency across the collection of documents.

The basic idea is:

TF-IDF = Term Frequency × Inverse Document Frequency

Words that are common throughout the dataset receive lower importance, while words that are more distinctive receive higher importance.

11.2 Advantages of TF-IDF

TF-IDF was suitable for the project because:

- It converts text into numerical features.
 - It captures important words.
 - It reduces the impact of extremely common terms.
 - It works effectively with traditional Machine Learning algorithms.
 - It is computationally practical for text classification.
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CHAPTER 12: MACHINE LEARNING MODEL DEVELOPMENT

Multiple Machine Learning algorithms were implemented and evaluated.

The models included:

1. Naive Bayes

2. Extra Trees
3. Random Forest
4. Support Vector Classifier
5. Logistic Regression
6. AdaBoost
7. XGBoost
8. Ensemble models

12.1 Naive Bayes

Naive Bayes is a probabilistic classification algorithm commonly used for text classification.

It is computationally efficient and can work effectively with TF-IDF features.

12.2 Random Forest

Random Forest combines multiple decision trees to make predictions.

It can capture different patterns in data and is generally more robust than an individual decision tree.

12.3 Extra Trees

Extra Trees is an ensemble-based tree algorithm that introduces additional randomness while creating decision trees.

It was included to compare tree-based classification performance.

12.4 Support Vector Classifier

Support Vector Classification attempts to identify an effective decision boundary between different classes.

SVC is particularly useful for high-dimensional feature spaces, making it suitable for TF-IDF-based text classification.

12.5 Logistic Regression

Logistic Regression is a widely used classification algorithm.

It provides a strong baseline for binary classification problems and works effectively with sparse text features.

12.6 AdaBoost

AdaBoost is an ensemble learning technique that combines multiple weak learners to produce a stronger classifier.

12.7 XGBoost

XGBoost is a gradient boosting algorithm designed for efficient and powerful predictive modelling.

CHAPTER 13: MODEL COMPARISON AND EVALUATION

Different models were trained and evaluated to determine which approach provided the most effective classification performance.

13.1 Evaluation Metrics

The major evaluation metrics considered were:

Accuracy

Accuracy measures the proportion of correctly classified messages.

Accuracy = Correct Predictions / Total Predictions

Precision

Precision measures how many messages predicted as spam were actually spam.

Precision = True Positives / (True Positives + False Positives)

Recall

Recall measures how many actual spam messages were successfully detected.

Recall = True Positives / (True Positives + False Negatives)

F1-Score

F1-score combines precision and recall into a single metric.

It is especially useful when class distributions are not perfectly balanced.

CHAPTER 14: ENSEMBLE LEARNING AND FINAL MODEL

After evaluating individual Machine Learning algorithms, ensemble learning techniques were considered.

14.1 Soft Voting Classifier

The best recorded result was achieved using a **Soft Voting Classifier** combining:

- Support Vector Classifier (SVC)
- Multinomial Naive Bayes
- Extra Trees

The ensemble approach combines predictions from multiple models rather than relying on only one classifier.

14.2 Final Results

The best recorded performance was:

Metric	Result
Accuracy	98.16%
Precision	99.17%

The results demonstrated that combining multiple classifiers could provide strong classification performance.

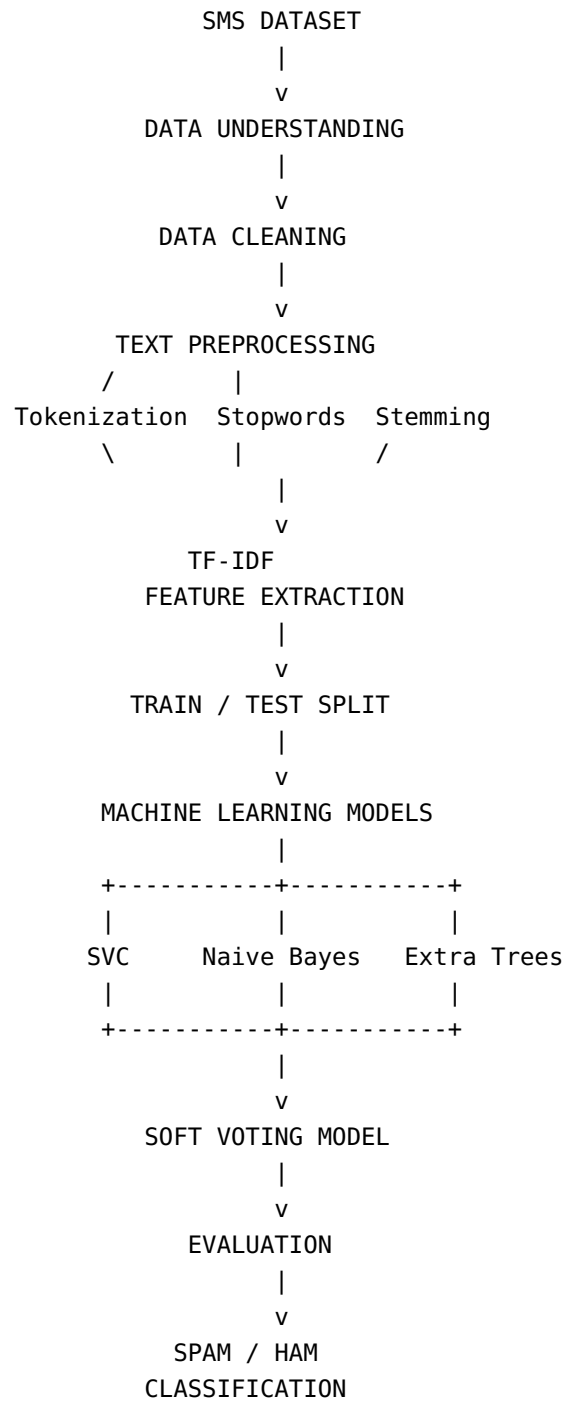
14.3 Significance of the Result

The high precision indicates that the model was effective at correctly identifying messages predicted as spam.

The ensemble approach also demonstrated the value of comparing different algorithms rather than selecting a model without experimentation.

CHAPTER 15: PROJECT WORKFLOW AND SYSTEM ARCHITECTURE

The overall project workflow can be represented as:



15.1 Development Process

The project followed a systematic development process:

Step 1 – Dataset Preparation

The dataset was loaded and inspected.

Step 2 – Data Cleaning

Duplicate and unnecessary records were removed.

Step 3 – Text Processing

SMS messages were cleaned, tokenized, and processed.

Step 4 – Feature Extraction

TF-IDF was applied to convert text into numerical vectors.

Step 5 – Model Training

Multiple Machine Learning algorithms were trained.

Step 6 – Model Evaluation

The models were compared using classification metrics.

Step 7 – Ensemble Learning

A Soft Voting Classifier was created using selected models.

Step 8 – Final Evaluation

The final model achieved the best recorded performance of **98.16% accuracy and 99.17% precision**.

CHAPTER 16: APPLICATIONS AND LIMITATIONS

16.1 Applications

The SMS spam detection system can be applied to:

- Mobile messaging applications.
- SMS filtering systems.
- Communication platforms.
- Fraud detection systems.
- Promotional message filtering.
- Automated message classification.

16.2 Limitations

Despite the strong results, the project has certain limitations.

Dataset Dependency

The model's performance depends on the quality and characteristics of the dataset used for training.

Changing Spam Patterns

Spam messages continuously change their language and patterns.

Language Limitations

The effectiveness of the model may vary for languages or writing styles not adequately represented in the training dataset.

Real-Time Deployment

A trained model requires additional engineering for large-scale real-time deployment.

CHAPTER 17: FUTURE SCOPE

The project can be further improved in several ways.

17.1 Larger Dataset

A larger and more diverse dataset can improve the model's ability to generalize.

17.2 Real-Time Application

The classifier can be integrated into a web or mobile application where users can enter messages and receive immediate predictions.

17.3 Advanced NLP

More advanced Natural Language Processing techniques can be explored.

Potential approaches include:

- Word embeddings.
- Deep Learning.
- Transformer-based models.

- Context-aware language models.

17.4 Multilingual Spam Detection

The system could be extended to support multiple languages.

17.5 Web-Based Interface

A user-friendly web interface could be developed where users enter an SMS message and the system displays whether it is:

HAM or **SPAM**

This would also connect the Machine Learning component with the **Web Development** focus of the internship.

CHAPTER 18: LEARNING OUTCOMES

The internship significantly improved my technical and professional skills.

18.1 Technical Learning

I gained practical experience in:

- Python programming.
- Data handling.
- Data preprocessing.
- Natural Language Processing.
- TF-IDF feature extraction.
- Machine Learning.
- Classification algorithms.
- Ensemble learning.
- Model evaluation.
- Data visualization.
- Project documentation.

18.2 Software Development Learning

As a Web Development Intern, I also developed a better understanding of:

- Structured programming.
- Project organization.
- Requirement understanding.
- Testing.

- Debugging.
- Development workflows.
- Documentation.
- Problem-solving.

18.3 Analytical Skills

Comparing multiple models helped me understand that selecting a model requires experimentation and evaluation.

Instead of assuming that one algorithm would always perform best, I learned to test different approaches and make decisions based on results.

CHAPTER 19: PROFESSIONAL DEVELOPMENT DURING INTERNSHIP

The internship was not limited to technical knowledge. It also helped develop professional skills.

19.1 Communication

Working in a professional environment helped improve the ability to communicate technical concepts clearly.

19.2 Time Management

The fixed internship duration required tasks to be planned and completed within appropriate timeframes.

19.3 Independent Learning

The project required learning different tools and concepts independently.

This strengthened my ability to research problems and understand unfamiliar technologies.

19.4 Team and Mentor Interaction

Interaction with project heads and mentors provided an understanding of professional feedback and project guidance.

19.5 Documentation

Preparing project documentation improved my ability to present technical work in a structured and understandable manner.

CHAPTER 20: CONCLUSION

The one-month internship at **Codec Technologies Pvt. Ltd.** was a valuable learning experience that provided practical exposure to software development and project implementation.

As a **Web Development Intern**, I gained an understanding of professional development practices, programming, debugging, documentation, and systematic problem-solving. The assigned project, **SMS Spam Detection Using Machine Learning**, further provided an opportunity to apply Python, Natural Language Processing, data preprocessing, TF-IDF, Machine Learning algorithms, and ensemble techniques to a real-world classification problem.

The project involved processing a dataset of 5,169 cleaned SMS messages and evaluating multiple Machine Learning algorithms. The best recorded result was achieved using a Soft Voting Classifier combining **SVC, Multinomial Naive Bayes, and Extra Trees**, achieving **98.16% accuracy and 99.17% precision**.

The internship helped strengthen both my technical and professional skills. More importantly, it gave me practical experience in approaching a project from data preparation and implementation to evaluation and documentation.

Overall, the internship has contributed significantly to my understanding of software development and has encouraged me to continue exploring areas such as **Web Development, Python, Data Analysis, Machine Learning, Natural Language Processing, Artificial Intelligence, and Software Development**.

CHAPTER 21: REFERENCES

1. Python Documentation
 2. Pandas Documentation
 3. NumPy Documentation
 4. Scikit-learn Documentation
 5. NLTK Documentation
 6. Matplotlib Documentation
 7. Seaborn Documentation
 8. Machine Learning and Natural Language Processing learning resources
 9. Internship training material provided during the internship
 10. SMS Spam Classification dataset and project resources
-

CHAPTER 22: APPENDIX

Appendix A – Internship Details

Organization: Codec Technologies Pvt. Ltd.

Role: Web Development Intern

Duration: 01 July 2026 – 30 July 2026

Mode: Pan India, Hybrid

Type: AICTE & ICAC Approved

Appendix B – Project Details

Project Title: SMS Spam Detection Using Machine Learning

Dataset Size Before Cleaning: 5,572 records

Dataset Size After Cleaning: 5,169 records

Ham Messages: 4,516

Spam Messages: 653

Feature Extraction: TF-IDF

Models Tested:

- Naive Bayes
- Extra Trees
- Random Forest
- SVC
- Logistic Regression
- AdaBoost
- XGBoost
- Ensemble Models

Best Model:

Soft Voting Classifier

Base Models:

- SVC
- Multinomial Naive Bayes
- Extra Trees

Best Accuracy: 98.16%

Best Precision: 99.17%

FINAL ACKNOWLEDGEMENT

The completion of this internship has provided me with meaningful practical experience and a better understanding of how technical knowledge can be applied to real-world development projects.

The experience gained as a **Web Development Intern at Codec Technologies Pvt. Ltd.** will serve as a strong foundation for my future academic projects, technical learning, and professional career.